

Enhancing Hospital Network Resilience Using Redundancy Protocols

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Abstract

Hospital networks require consistently high availability and reliability to support critical healthcare services; network failures may disrupt clinical systems and directly compromise patient safety. This study aimed to enhance the resilience of an existing hospital network design, originally proposed for Medicine City Hospital by Jassim (2024) by implementing redundancy protocols within a simulated two-tier, collapsed-core enterprise network environment using Cisco Packet Tracer version 9.0.0. The baseline network implemented Virtual Local Area Networks (VLANs) and Open Shortest Path First (OSPF) routing but lacked any form of gateway or link redundancy, rendering it vulnerable to single points of failure. The network was augmented with Hot Standby Router Protocol (HSRP) for default gateway redundancy and Rapid Spanning Tree Protocol (RSTP) for fast Layer 2 convergence and loop prevention. A peer link was also introduced between the two Multilayer Switches (MLS) on both the Headquarters and Branch sides. Network performance was evaluated under controlled failure scenarios that simulate both device failure and link failure based on four metrics: availability, failover time, packet loss and packet delivery ratio (PDR). Results demonstrate that the proposed redundancy mechanisms substantially improve network availability, reduce failover time, lower packet loss and achieve a higher PDR relative to the baseline configuration, thereby validating their effectiveness in enhancing hospital network resilience. The findings provide practical reference for real-world hospital network deployments where continuous connectivity is a clinical necessity.

Keywords

Hospital network, HSRP, Network redundancy, RSTP, Network resilience, Network availability, Cisco Packet Tracer.

1. Introduction

Modern healthcare facilities depend on interconnected digital systems including Electronic Health Record (EHR) platforms, laboratory interfaces, medical imaging equipment and real-time communication tools to deliver timely and accurate patient care [1]. The continuous availability of these systems is therefore not merely a technical concern but a direct determinant of patient outcomes. Network failures in hospital environments can prevent clinical staff from accessing patient records, stall diagnostic workflows, and, in critical scenarios, endanger lives [2]. Studies indicate that healthcare organizations experience between four and six network downtime incidents annually, with each lasting an average of 2.5 to 4 hours [3].

High availability in network infrastructure is conventionally measured using the 'nines of availability' framework as shown in *Table 1.1*, in which 99.999% (Five Nines) represents the gold standard, permitting no more than approximately 5.26 minutes of downtime per year [4]. Achieving this level of availability in a hospital environment necessitates the strategic deployment of network redundancy, which is the practice of provisioning alternative devices, links and paths that automatically assume active roles upon the failure of primary components [5].

Two protocol families are especially relevant to this goal. First Hop Redundancy Protocols (FHRPs) such as Hot Standby Router Protocol (HSRP), Virtual Router Redundancy Protocol (VRRP) and Gateway Load Balancing Protocol (GLBP) to address the problem of gateway availability by virtualizing multiple routers or multilayer switches into a single logical gateway endpoint [6]. At the data-link layer, Rapid Spanning Tree Protocol (RSTP, IEEE 802.1w) significantly reduces convergence time relative to the traditional Spanning Tree Protocol (STP), transitioning backup ports to a forwarding state in seconds rather than the 30–50 seconds characteristic of STP [7]. Together, these protocols form complementary layers of redundancy spanning the network hierarchy.

The baseline network adopted in this study is a smart hospital enterprise design proposed by Jassim (2024) for Medicine City Hospital, implemented in Cisco Packet Tracer as shown in *Figure 1.1* [8]. While the design incorporates VLANs, OSPF, DHCP, SSH, ACLs, NAT and IPSec VPN, it contains no gateway or link redundancy mechanisms. This absence creates

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a critical vulnerability: a single Multilayer Switch or uplink failure can render all downstream VLANs and by extension, the EHR system completely inaccessible.

Availability	Downtime / Year	Downtime / Month	Downtime / Week	Downtime / Day
99.999%	5.256 minutes	0.438 minutes	0.101 minutes	0.014 minutes
99.995%	26.28 minutes	2.19 minutes	0.505 minutes	0.072 minutes
99.990%	52.56 minutes	4.39 minutes	1.011 minutes	0.144 minutes
99.950%	4.38 hours	21.9 minutes	5.054 minutes	0.72 minutes
99.900%	8.76 hours	43.8 minutes	10.108 minutes	1.44 minutes
99.500%	43.8 hours	3.85 hours	50.538 minutes	7.2 minutes
99.250%	65.7 hours	5.475 hours	75.808 minutes	10.8 minutes
99.000%	87.6 hours	7.3 hours	101.077 minutes	14.4 minutes

Table 1.1: "Nines of Availability" [4]

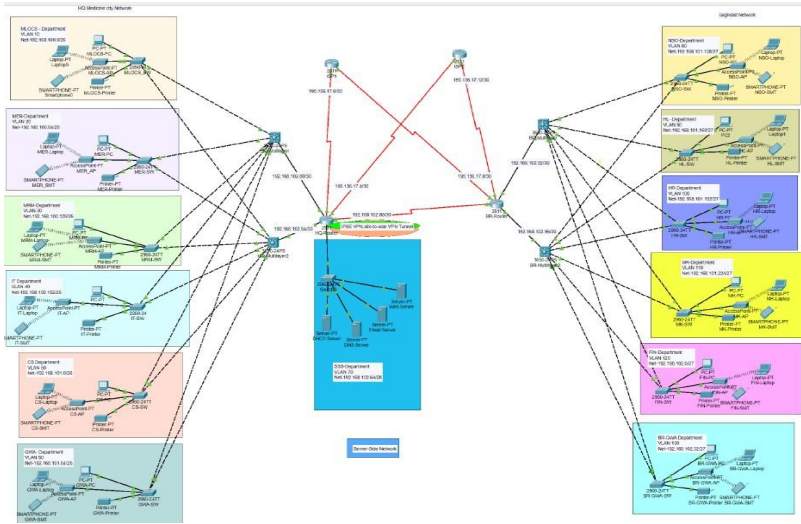


Figure 1.1: Design of a hospital network from Smart Hospital Network Enterprise Design for Medicine City Hospital via Packet Tracer by Mohammed Rajih Jassim [8]

This study addresses that gap with three principal objectives: (1) to examine the theoretical role of redundancy protocols in improving network resilience and availability within a hospital environment; (2) to develop and simulate an enhanced hospital network implementing HSRP for gateway redundancy and RSTP for rapid Layer 2 convergence within Cisco Packet Tracer; and (3) to quantitatively compare the performance of the enhanced network against the baseline design using four metrics — network availability, failover time, packet loss and PDR.

The remainder of this paper is structured as follows. Section 2 reviews related literature on redundancy protocols and hospital network performance. Section 3 describes the materials and methods employed. Section 4 presents and

discusses the results. Section 5 concludes the study.

2. Related Work

2.1 Hospital Network Infrastructure and Resilience

Rodrigues et al. [9] modelled the availability of a smart hospital architecture using Stochastic Petri Nets (SPN) and sensitivity analysis. Their results showed that a server configuration with full redundancy achieved the highest availability of 99.9199%, corresponding to approximately 7 hours of downtime per year, approaching the Five Nines benchmark. The study underlines that high availability is a prerequisite for quality of service in healthcare systems and that redundancy is the principal enabler of that availability.

A key limitation, however, is that the model relied on synthetic validation data and low-capacity test machines, which may not reflect the scale of real hospital deployments.

Jassim [8] proposed a comprehensive smart hospital enterprise network design for Medicine City Hospital using Cisco Packet Tracer, incorporating VLANs for departmental segmentation, OSPF for dynamic routing, DHCP for IP address management, SSH for secure remote management, ACLs and NAT for security and address translation, and an IPSec Site-to-Site VPN for inter-site connectivity. While the design demonstrates a well-structured enterprise network, it explicitly lacks redundancy mechanisms, which this study directly addresses.

2.2 First Hop Redundancy Protocol

Shahriar and Fan [6] conducted a comparative performance analysis of the three FHRP family members: HSRP, VRRP and GLBP, in a multi-VLAN network with STP using GNS3 and Wireshark. Their results showed that HSRP achieved the lowest average time consumption in *Table 2.1* (57.3 s under normal conditions, 86.6 s under link failure), the highest average packet transmission rate in *Table 2.2* (44.25 packets/s normal, 28.75 packets/s under failure), and the lowest end-to-end delay of the three protocols in *Figure 2.1*. They concluded that HSRP is the recommended FHRP for small-to-medium organizations with complex VLAN environments. A noted limitation is that failure scenarios were confined to link failures, with no analysis of complete node failure.

Rachman Harahap and Handoko [10] implemented HSRP as a Disaster Recovery Plan (DRP) mechanism in a simulated network connecting a Network Operation Centre to an Operations Maintenance division. The HSRP Active router successfully transferred traffic to the Standby router upon failure, demonstrating seamless failover with zero packet loss in the simulated environment. The study confirms HSRP's ability to determine the best available path during a primary link failure. However, the results were entirely simulation-based using GNS3 and Cisco Packet Tracer, limiting generalization to physical deployments.

Firdaus Syah and Adriyanto [11] compared HSRP and VRRP in an enterprise core network simulation using Cisco Packet Tracer, quantifying failover times and availability improvements. Their findings confirm that HSRP exhibits faster failover speeds than VRRP, while VRRP offers better cross-vendor compatibility and stability due to its open-standard nature. Rasyid et al. [12] validated HSRP in a school network simulation, reporting zero packet loss during failover scenarios when measured against THIPON QoS standards, though noting a slight throughput reduction during the recovery phase compared to normal operation.

Simanjuntak et al. [13] applied HSRP in a banking sector network topology with core, distribution, and access layers, simulated using Cisco Packet Tracer and GNS3. Their results confirmed that HSRP successfully minimized link downtime by automatically switching to the standby router, preventing transaction disruptions. The study also measured throughput, delay, jitter and packet loss, providing a comprehensive QoS profile of HSRP behavior during failure events.

2.3 Layer 2 Redundancy and RSTP

Rathod [7] compared STP and RSTP in a five-switch ring topology using Cisco Packet Tracer, specifically measuring packet loss during convergence. RSTP lost only one packet (7% loss) during a link failure event, compared to twelve packets (53% loss) for STP as shown in *Table 2.3*. The study demonstrated that both protocols provide backup path redundancy by unblocking a previously blocked port upon link failure, but RSTP does so with substantially less disruption to live traffic. The limitation of the study is its reliance on a simple ring topology, which may not capture the complexity of real enterprise environments.

Anggi Silalahi et al. [14] implemented and compared STP and RSTP in Cisco Packet Tracer using a three-switch topology with forced redundant paths. They observed that traditional STP required approximately 30 seconds to converge, passing through Listening and Learning states, while RSTP converged in a matter of seconds by transitioning directly to the Forwarding state as shown in *Table 2.4*. The study confirms that convergence time is proportional to failover time, reinforcing the operational advantage of RSTP in environments requiring rapid recovery.

First Redundancy Protocol (FHRP)	Hop	Average Consumed Time For no Link Failure (sec)	Average Consumed Time For a Failed Link (sec)
HSRP		57.3	86.6
VRRP		60.1	90
GLBP		61.5	91.5

Table 2.1: Average Time Consumption [6]

First Redundancy Protocol (FHRP)	Hop	Average Packet/1sec For no Link Failure	Average Packet/1sec For a Failed Link
HSRP		44.25	28.75
VRRP		43.7	24.35
GLBP		42.15	23.96

Table 2.2: Average Packet Transmission Rate [6]



Figure 6. End to end delay performance analysis of FHRP when there is no link failure.



Figure 7. End to end delay performance analysis of FHRP for link failure.

Figure 2.2:End-to-End Delay [6]

Parameters	STP	RSTP
Standard	IEEE 802.1w	IEEE 802.1D
Number of Packet Sent	23	16
Number of Packet Received	11	15
Number of Packet Lost	12	1
Loss	53%	7%

Table 2.3: STP and RSTP Packet Loss Comparison Table [7]

Fitur	STP (802.1D)	RSTP (802.1w)
Nama Lengkap	Spanning Tree Protocol	Rapid Spanning Tree Protocol
Protokol di output	Protocol ieee	Protocol rstp
Kecepatan konvergensi	Lambat (hingga 30 detik)	Cepat,dalam beberapa detik
Transisi port	Listening-Learning-Forwarding	Langsung ke Forwarding

Table 2.4: STP and RSTP comparison table [14]

2.4 Combined Redundancy Implementation

Octavian [16] designed a redundant network for PLN UID Jakarta Raya using EtherChannel, HSRP and inter-VLAN routing in Cisco Packet Tracer. When the primary multilayer switch was disabled, the standby switch seamlessly assumed the active role, demonstrating the effectiveness of combining HSRP gateway redundancy with EtherChannel link aggregation to eliminate single points of failure. The study is, however, tailored to a specific utility company topology and may require modification for direct application to healthcare environments.

Yogasetya Suhandi et al. [17] designed an HSRP-based network infrastructure for PT. Quantum using Cisco Packet Tracer, reporting zero packet loss during Active-to-Standby transitions and latency consistently below 10 ms. Alfian Armawan Sandi et al. [18] demonstrated that LACP-based link aggregation in combination with VLANs significantly reduced file transfer times and improved network stability, though at the cost of higher CPU utilization.

2.5 Summary and Research Gap

The literature consistently demonstrates that HSRP and RSTP are effective redundancy mechanisms across enterprise and institutional network environments, improving availability, reducing failover time and minimizing packet loss. However, studies focusing specifically on the combined deployment of HSRP and RSTP within a hospital network topology and their quantitative impact on all four-performance metrics simultaneously remain scarce. Furthermore, most existing studies either address a single protocol or evaluate redundancy in non-healthcare contexts such as banking, manufacturing or educational institutions. This study directly addresses this gap by implementing both protocols in a simulated hospital network and rigorously comparing their performance impact against a documented baseline using controlled failure scenarios.

3. Materials and Methods

3.1 Simulation Environment

All network design, configuration and testing activities were conducted within Cisco Packet Tracer version 9.0.0, a discrete-event network simulator developed by Cisco Systems. This platform supports specific device types (Cisco 3650 Multilayer Switches, Cisco 2960 Layer 2 Access Switches, and Cisco ISR routers), inter-VLAN routing configurations and redundancy protocol syntax required by this study. Network architecture and framework diagrams were produced using Canva. No physical hardware was employed, and all

results are therefore derived from simulated traffic under simulated failure conditions.

3.2 System Development Lifecycle Model

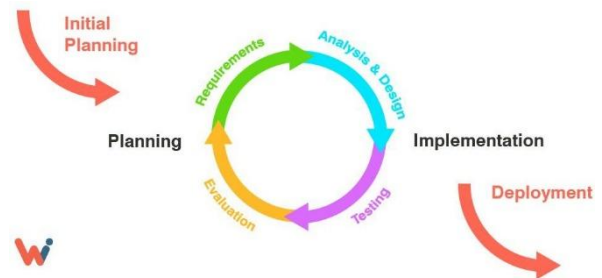


Figure 3.1: Iterative Model

The project followed an iterative System Development Life Cycle (SDLC) model comprising six phases: requirement analysis, network design, implementation, testing, deployment as well as maintenance and evaluation. The iterative approach enabled successive cycles of baseline network replication, failure testing, redundancy enhancement and re-testing, thereby supporting controlled comparison between configurations.

In Phase 1 (Requirement Analysis), the baseline paper was reviewed to identify the topology, protocols and configurations, and the absence of redundancy mechanisms was identified as the principal limitation. In Phase 2 (Network Design), the original network was replicated in Cisco Packet Tracer. Phase 3 (Implementation) comprised two stages: first, replicating and configuring the baseline network; and second, augmenting it with HSRP and RSTP. Phase 4 (Testing) captured performance metrics under normal and failure conditions for both the baseline and enhanced networks. Phase 5 (Deployment) treats the enhanced configuration as the reference deployment. Phase 6 (Maintenance and Evaluation) involved comparative analysis and documentation of limitations.

3.3 Network Architecture

The study adopted a two-tier collapsed-core architecture as shown in *Figure 3.2* and *Figure 3.3*, in which the core and distribution layers are merged into a single tier, reducing hardware complexity while retaining inter-VLAN routing and redundancy capabilities. The upper (collapsed core/distribution) tier comprises two Multilayer Switches (MLS1 and MLS2) on both the Headquarters (HQ) and Branch sides, connected to their respective routers and ISP links. The lower (access) tier consists of Layer 2 Access Switches, each assigned to a specific departmental VLAN.

The complete network comprises 13 VLANs:

six on the HQ side (VLAN 10-60), six on the Branch side (VLAN 80-130), and one server VLAN (VLAN 70) hosting DHCP, DNS, Email, and Web servers. A Management VLAN (VLAN 99) and an unused-port security VLAN (VLAN 999) are configured across all switches.

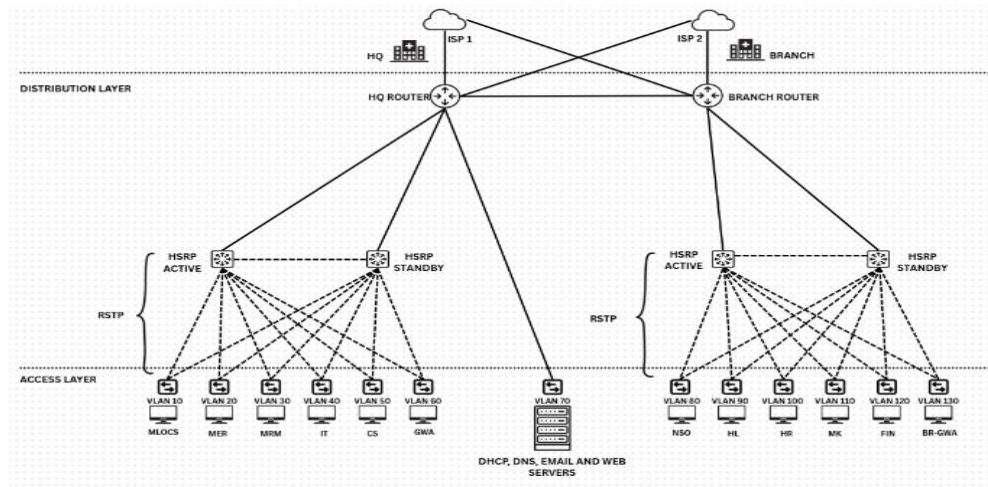


Figure 3.3: Network Architecture

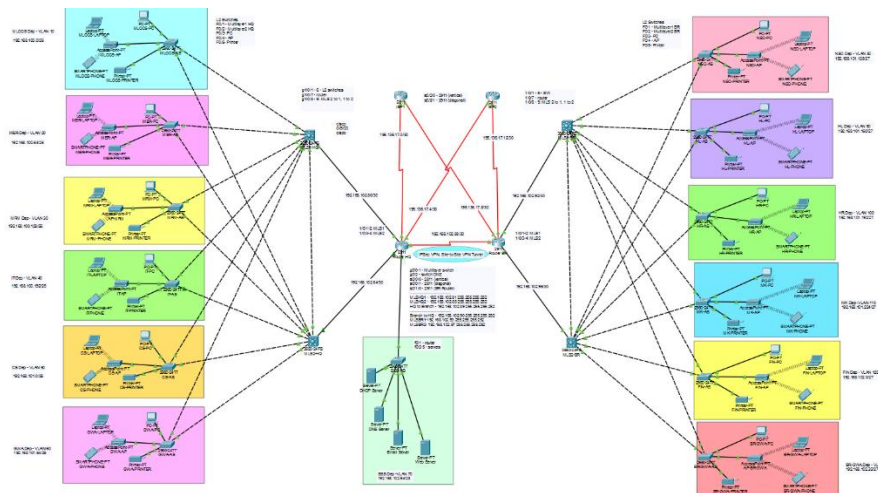


Figure 3.3: Network Architecture

3.4 Network Architecture

The baseline network replicates the original configuration from Jassim [8]. Configured protocols in the baseline include OSPF for inter-site and inter-VLAN routing, DHCP for dynamic IP address assignment. STP was present as the default spanning-tree mode on the switches, but no HSRP nor MLS peer link were configured. This baseline configuration was tested first to establish reference performance values under both normal and failure conditions.

3.5 Enhanced Network Configuration

The enhanced network retains the baseline configurations and adds the following redundancy

mechanisms. HSRP was configured on a per-VLAN basis on MLS1 and MLS2 on both HQ and Branch sides. MLS1 was assigned a higher HSRP priority to ensure it assumes the Active role under normal conditions, and pre-emption was enabled to allow MLS1 to automatically reclaim the Active role after recovery. A virtual IP address was configured for each VLAN and designated as the default gateway for all end devices in that VLAN. RSTP (IEEE 802.1w) was verified and enforced on all MLS and Access Switch trunk interfaces, and a dedicated peer link was introduced between MLS1 and MLS2 to provide an alternate Layer 2 forwarding path in the event of an uplink failure.

3.6 HSRP Operation

HSRP operates, as shown in *Figure 3.4*, by grouping two or more routers or multilayer switches into a virtual router group, presenting a single virtual IP and MAC address to end devices. The Active switch forwards traffic for the group while the Standby switch monitors the Active switch through periodic Hello messages exchanged every 3 seconds (default). If the Standby's hold timer expires, signaling that the Active switch has stopped sending Hello messages, the Standby immediately promotes itself, assumes the virtual IP and MAC address and resumes traffic forwarding without

requiring any reconfiguration of end-device gateways.

Upon recovery of the previously failed Active switch, the pre-empt feature forces a priority comparison. If the recovered switch holds a higher priority than the current Active switch, it waits for the pre-empt delay timer (to ensure full stabilization) before reclaiming the Active role as shown in *Figure 3.5*.

This mechanism prevents unnecessary role oscillation caused by rapid device restarts.

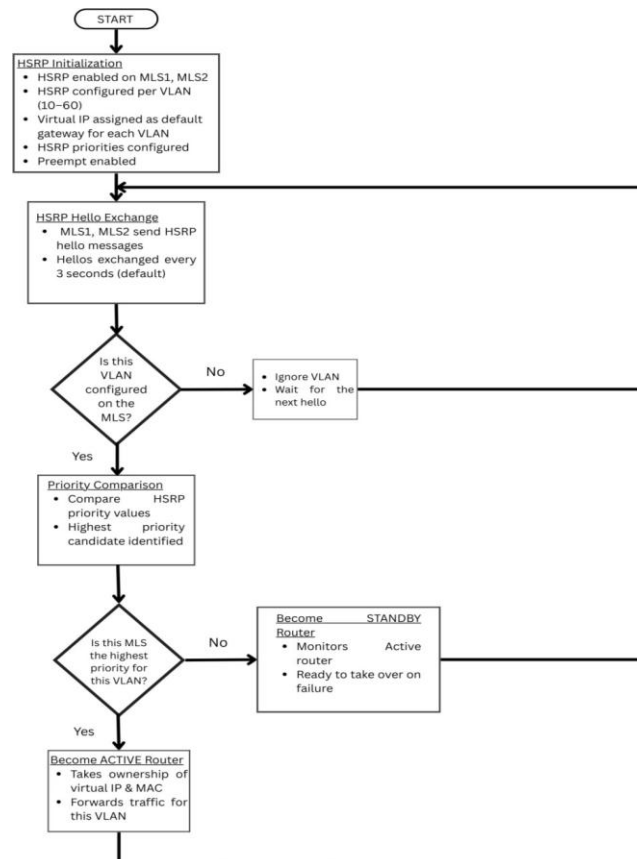


Figure 3.4: HSRP Role Assignment

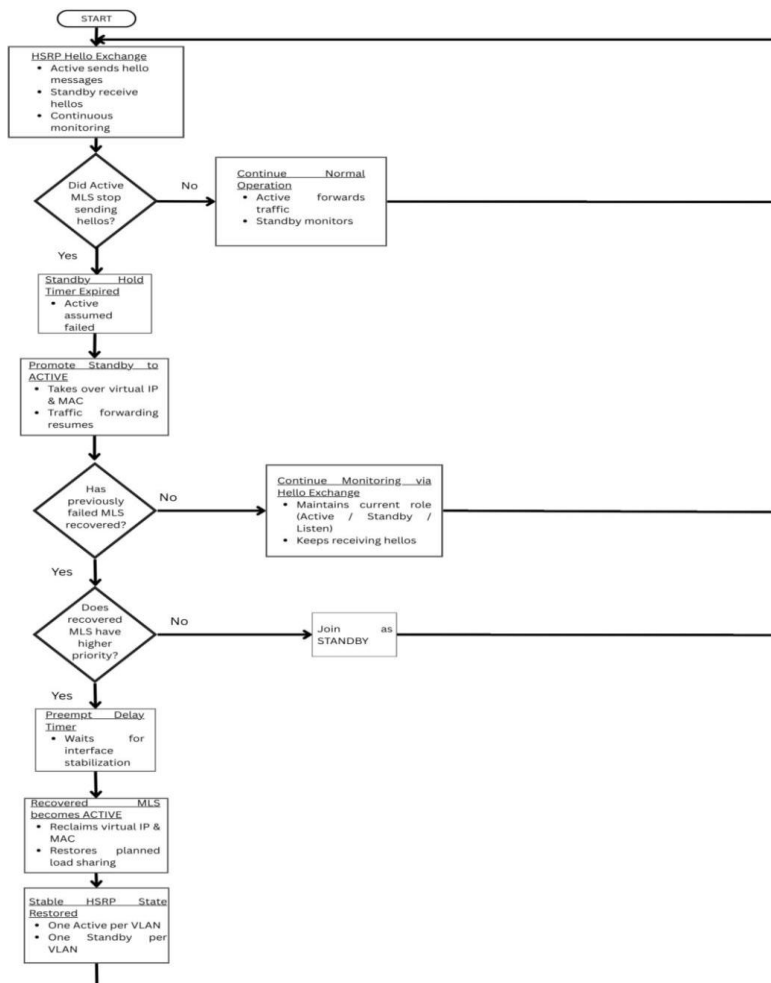


Figure 3.5: HSRP Failover + Recovery

3.7 RSTP Operation

RSTP prevents Layer 2 broadcast storms which can occur when multiple redundant paths exist in a switched network by electing a Root Bridge and assigning each switch port a role: Root Port, Designated Port, or Alternate/Backup Port. Alternate ports are maintained in a Blocking/Discarding state under normal conditions, ready to transition to Forwarding state almost instantaneously upon detection of a topology change. Unlike STP, which requires passage through Listening and Learning states (totaling approximately 30 seconds), RSTP uses a proposal-and-agreement handshake mechanism that enables convergence within seconds, minimizing the window during which traffic is disrupted. *Figure 3.6* shows the flow of RSTP Process.

3.8 Failure Scenario

Testing was conducted under two controlled failure scenarios applied to both the baseline and enhanced networks. In Scenario A (device failure), MLS1 was administratively shut down to simulate complete device failure, assessing HSRP gateway failover behavior. In Scenario B (link failure), the uplink connecting MLS1 to its downstream Layer 2 Access Switches was disabled, assessing RSTP convergence and backup path activation. In both scenarios, a continuous ping was initiated between a host in one VLAN and a host in a separate VLAN for a total observation period of 100 seconds. Testing was performed exclusively on the HQ infrastructure, as the HQ and Branch sides share identical structural configurations.

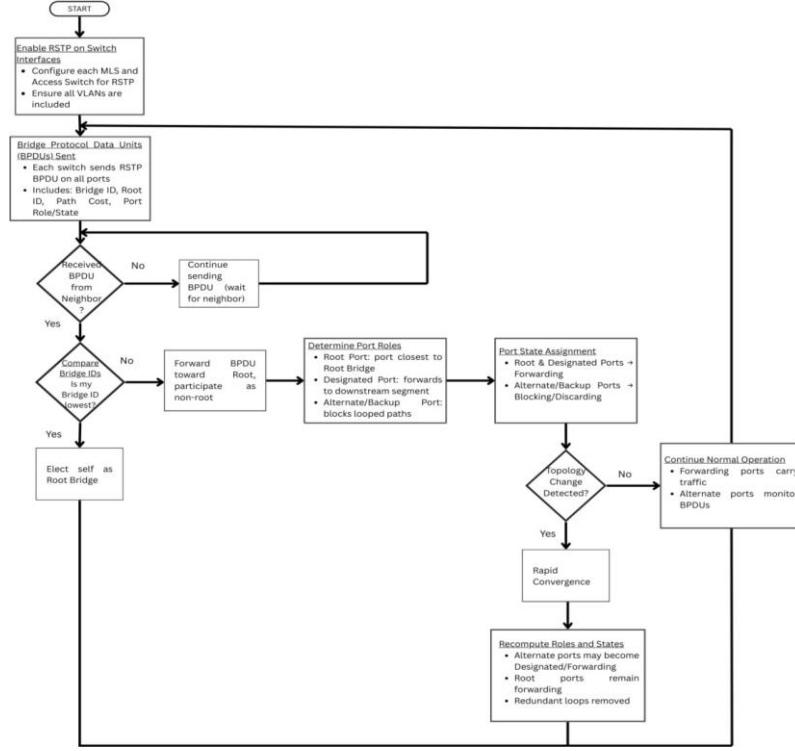


Figure 3.6: RSTP Process

3.9 Performance Metrics and Formulas

The Four quantitative performance metrics were used to evaluate network behavior before and after redundancy implementation.

Failover Time is the duration between the onset of a failure and the resumption of successful packet delivery, determined by observing continuous ping output:

$$\text{Failover Time} = T_{\text{ping resumes}} - T_{\text{failure}}$$

Availability represents the proportion of the 100-second observation window during which the network remained operational:

$$\text{Availability}(\%) = \left(\frac{\text{Total Time} - \text{Downtime}}{\text{Total Time}} \right) \times 100$$

Where:

- Total Time = 100 s
- Downtime = Measured Failover Time.

Packet Loss is the number of packets that failed to reach their destination across the complete ping session, including the failure and recovery period:

$$\text{Packet Loss} = \text{Packet Sent} - \text{Packet Received}$$

Packet Delivery Ratio (PDR) expresses the proportion of transmitted packets successfully received:

$$\text{PDR}(\%) = \left(\frac{(\text{Packet Sent} - \text{Packet Loss})}{\text{Packet Sent}} \right) \times 100$$

4. Results

4.1 Overview

Performance was evaluated across four controlled failure scenarios: Scenario 1, link failure without the MLS peer link; Scenario 2, link failure with the peer link active; Scenario 3, active MLS (MLS1) failure without the peer link; and Scenario 4, active MLS failure with the peer link active. Each scenario was tested under two protocol configurations: HSRP combined with legacy STP and HSRP combined with RSTP and, where applicable, compared against a baseline in which no FHRP was deployed. Testing was performed exclusively on the HQ infrastructure, as the HQ and Branch sides share identical structural configurations. In all tests, 50 packets were issued over a 100-second observation window between hosts in different VLANs. The four metrics recorded — availability, failover time, packet loss and Packet Delivery Ratio (PDR) were computed using the formulae defined in Section 3.9.

4.2 Scenario 1: Link failure without peer link

In Scenario 1, a link between MLS1 and its downstream access switches was deleted to simulate an uplink failure, with no peer link present between MLS1 and MLS2. Without any FHRP configured, the network could not recover automatically, requiring manual gateway reconfiguration before any traffic could resume. This establishes the critical baseline vulnerability of the original network design.

When HSRP was added alongside legacy STP, the network recovered but with a failover time of 25 s, 4 packets lost, 75% availability and 92% PDR. The prolonged recovery reflects STP's convergence behavior, which passes through Listening and Learning states before unblocking an alternate path, a process that can take up to 30 seconds. Replacing STP with RSTP reduced failover time to 7 s, cut packet loss to 1, raised availability to 93% and PDR to 98%. RSTP's proposal-and-agreement mechanism transitions blocked ports directly to the Forwarding state upon detecting a topology change, substantially narrowing the disruption window. These results are summarized in *Table 4.1*.

4.3 Scenario 2: Link failure with peer link

Scenario 2 repeated the link failure condition with the dedicated MLS peer link activated. For HSRP + RSTP, the peer link provided an immediate Layer 2 transit path that RSTP could exploit the instant the primary uplink failed. The result was a sub-second failover time (measured at 1 s, rounded from the Packet Tracer timer resolution), zero packet loss, approximately 99% availability and 100% PDR, which is the best outcome recorded across all scenarios. This combination effectively eliminates perceptible service disruption during link failure events.

By contrast, HSRP + STP with the peer link performed worse than in Scenario 1. The additional inter-switch link increased Layer 2 topology complexity, requiring STP to re-evaluate port roles and recalculate path costs before converging. This extended reconvergence raised failover time to 31 s, increased packet loss to 5, and reduced availability to 69% and PDR to 90%. This finding demonstrates that introducing a peer link without upgrading to RSTP is counterproductive and can actively degrade network availability relative to a simpler single-uplink topology. Results are presented in *Table 4.2*.

Protocol Combination	Availability (%)	Failover Time (s)	Packet Loss (Sent = 50)	PDR (%)
HSRP + STP	75%	25 s	4	92%
HSRP + RSTP	93%	7 s	1	98%
No HSRP	N/A — manual intervention required			

Table 4.1: Performance Metrics: Link Failure (Without Peer Link)

Protocol Combination	Availability (%)	Failover Time (s)	Packet Loss (Sent = 50)	PDR (%)
HSRP + STP	69%	31 s	5	90%
HSRP + RSTP	~99%	< 1 s	0	100%

Table 4.2: Performance Metrics: Link Failure (With Peer Link)

4.4 Scenario 3: Active MLS failure without peer link

Scenario 3 simulated a complete device failure by removing the power supply from MLS1 (the HSRP Active switch), without the peer link present. Both HSRP + STP and HSRP + RSTP produced identical results: 88% availability, a 12-second failover time, 2 packets lost and 96% PDR. This parity arises because the failover mechanism in a device failure is governed entirely by HSRP's hold timer. MLS2 (the Standby) must detect three consecutive missed Hello messages before promoting itself to the Active role; with a default Hello interval of 3 seconds and hold time of 10 seconds, this process takes approximately 10–12 seconds regardless of the Layer 2 spanning tree protocol in use. Since no uplink topology change occurs, RSTP's faster convergence algorithm provides no advantage over STP in this scenario. *Table 4.3* presents the results.

4.5 Scenario 4: Active MLS failure with peer link

Scenario 4 repeated the active MLS failure with the peer link present. HSRP + RSTP again recorded 88% availability, 12-second failover, 2 packets lost and 96% PDR, identical to Scenario 3, confirming that the peer link introduces no additional delay when RSTP is in use, since RSTP handles the added topology complexity seamlessly.

HSRP + STP, however, degraded considerably. With MLS1 removed, STP had to re-elect port roles

and recalculate path costs across the more complex topology that includes the peer link. This reconvergence was substantially slower than in Scenario 3, driving failover time up to 30 s, packet loss to 5 and availability down to 70%, with PDR falling to 90%. This result further underlines that STP's inability to efficiently process additional inter-switch paths makes it an unsuitable companion to a peer-link architecture. *Table 4.4* summarises these results.

4.6 Summary of the Results

Table 4.5 consolidates all performance metrics across the four scenarios. HSRP + RSTP consistently outperforms or matches HSRP + STP in every tested condition. The most notable gains occur in Scenario 2, where the combination of the peer link with RSTP produces near-zero failover and perfect PDR. When failover is determined by the HSRP hold timer rather than Layer 2 convergence speed (Scenarios 3 and 4 without peer link), both configurations perform identically, confirming that HSRP timer expiry is the primary bottleneck during device-level failures. The peer link amplifies the performance differential between the two protocol combinations: it dramatically improves HSRP + RSTP outcomes during link failures but introduces reconvergence penalties for HSRP + STP during both link and device failures. These results collectively validate the effectiveness of the proposed redundancy mechanisms in enhancing hospital network resilience.

Protocol Combination	Availability (%)	Failover Time (s)	Packet Loss (Sent = 50)	PDR (%)
HSRP + STP	88%	12 s	2	96%
HSRP + RSTP	88%	12 s	2	96%

Table 4.3: Performance Metrics: Active MLS Failure (Without Peer Link)

Protocol Combination	Availability (%)	Failover Time (s)	Packet Loss (Sent = 50)	PDR (%)
HSRP + STP	70%	30 s	5	90%
HSRP + RSTP	88%	12 s	2	96%

Table 4.4: Performance Metrics: Active MLS Failure (With Peer Link)

Failure Scenario	Protocol Config	Availability (%)	Failover Time	Packet Loss	PDR (%)
Link Failure (No Peer Link)	HSRP + STP	75%	25 s	4 / 50	92%
Link Failure (No Peer Link)	HSRP + RSTP	93%	7 s	1 / 50	98%
Link Failure (With Peer Link)	HSRP + STP	69%	31 s	5 / 50	90%
Link Failure (With Peer Link)	HSRP + RSTP	~99%	< 1 s	0 / 50	100%
Active MLS Failure (No Peer Link)	HSRP + STP	88%	12 s	2 / 50	96%
Active MLS Failure (No Peer Link)	HSRP + RSTP	88%	12 s	2 / 50	96%
Active MLS Failure (With Peer Link)	HSRP + STP	70%	30 s	5 / 50	90%
Active MLS Failure (With Peer Link)	HSRP + RSTP	88%	12 s	2 / 50	96%

Table 4.5: Summary of All Performance Metrics Across All Failure Scenarios

Figure 4.1 — Availability (%): HSRP + RSTP achieves higher availability than HSRP + STP in all scenarios except Scenario 3, where both configurations are equal. The largest gap occurs in Scenario 2, where the peer link raises RSTP-based availability to approximately 99% while STP-based availability drops to 69%.

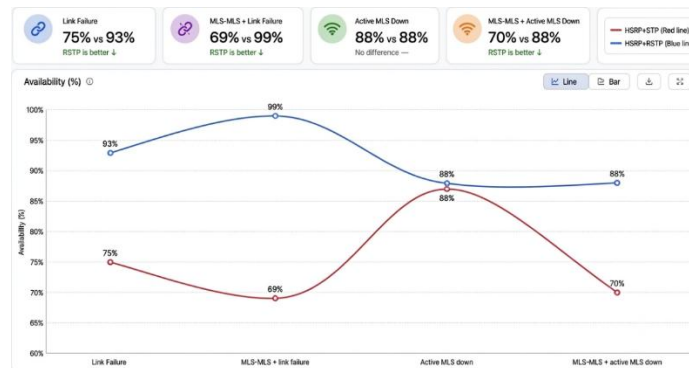


Figure 4.1: Availability Graph

Figure 4.2 — Failover Time (s): HSRP + RSTP consistently converges faster than HSRP + STP. The most significant difference is in Scenario 2, where RSTP achieves sub-second failover against STP's 31 seconds. Both configurations record identical 12-second failover in Scenarios 3 and 4 (without peer link), confirming HSRP timer expiry as the limiting factor during device failures.

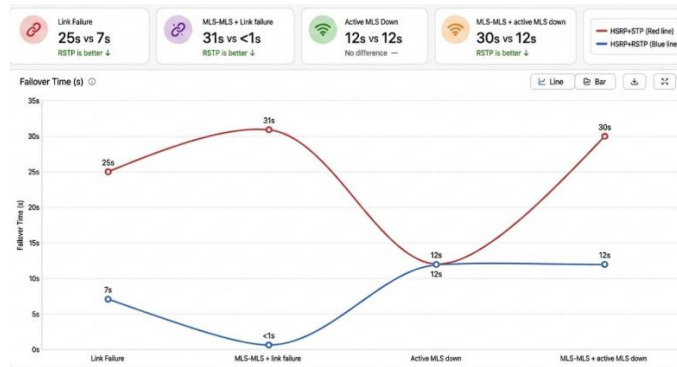


Figure 4.2: Failover Time Graph

Figure 4.3 — Packet Loss: HSRP + RSTP records equal or lower packet loss than HSRP + STP across all scenarios, reaching zero loss in Scenario 2. HSRP + STP peaks at 5 packets lost in Scenarios 2 and 4, both cases where the peer link is active.



Figure 4.3: Packet Loss Graph

Figure 4.4 — Packet Delivery Ratio (%): HSRP + RSTP maintains a higher or equal PDR compared to HSRP + STP in all scenarios, peaking at 100% in Scenario 2. The narrowest gap occurs in Scenarios 3 and 4 without the peer link, where both configurations achieve 96% PDR.



Figure 4.4: PDR Graph

5. Discussion

5.1 RSTP as the critical complement to HSRP

The results consistently demonstrate that RSTP is not merely an incremental upgrade to STP but a fundamentally different convergence mechanism whose advantages are decisive in any topology that includes redundant inter-switch paths. In Scenario 1, HSRP + RSTP reduced failover time by 72% relative to HSRP + STP (7 s versus 25 s) without any additional hardware, achieving this improvement solely through RSTP's proposal-and-agreement handshake, which bypasses the Listening and Learning delay states that cause STP to hold traffic for up to 30 seconds. This finding aligns with Rathod [7], who reported a 46-percentage-point reduction in packet loss when replacing STP with RSTP in a five-switch ring topology, and with Anggi Silalahi et al. [14], who observed STP convergence taking approximately 30 seconds compared to seconds for RSTP under identical failure conditions.

The advantage of RSTP is further amplified when the peer link is introduced. Scenario 2 showed HSRP + RSTP achieving near-perfect recovery (approximately 99% availability, zero packet loss, 100% PDR) precisely because RSTP could exploit the peer link as an immediate transit path the instant the primary uplink failed. This result surpasses the zero-packet-loss outcomes reported by Rachman Harahap and Handoko [10] and Yogasetya Suhandi et al. [17] in simpler single-VLAN HSRP deployments and extends them to a more complex multi-VLAN environment with a dedicated inter-switch failover path.

5.2 The peer link as a double-edged design variable

A significant and practically important finding is that the peer link between MLS1 and MLS2 does not unconditionally improve network performance, but instead, its effect is contingent on the spanning tree protocol in use. With RSTP, the peer link produced the best results observed in this study. With STP, however, the peer link worsened performance in both failure categories. In Scenario 2 (link failure with peer link), HSRP + STP availability fell to 69% with a 31-second failover — worse than the 75% availability and 25-second failover recorded in Scenario 1 without the peer link. Similarly, in Scenario 4 (device failure with peer link), HSRP + STP degraded from 88% to 70% availability and from 12 to 30 seconds failover time compared to Scenario 3.

This degradation occurs because each additional inter-switch link increases the number of port state transitions that STP must evaluate before converging. STP processes these transitions sequentially through Listening, Learning and Forwarding states, so topological complexity directly prolongs convergence time. RSTP's per-port proposal-and-agreement mechanism resolves each port independently and in parallel, making it insensitive to this added complexity. This interaction between the peer link and the spanning tree protocol is a design variable that is not consistently highlighted in existing literature and represents a practical risk: network engineers who deploy a peer link with the expectation of improved resilience may inadvertently worsen availability if STP remains in use. The implication for hospital network design is unambiguous — a peer link must be paired with RSTP to be beneficial.

5.3 HSRP hold timer as the device-failure bottleneck

Scenarios 3 and 4 revealed that when MLS1 is completely powered off, both HSRP + STP and HSRP + RSTP (without the peer link degrading STP) converge to an identical 12-second failover time. This parity confirms that the limiting factor in device-level failures is the HSRP hold timer, not Layer 2 convergence speed. With default settings, Hello interval 3 seconds, hold time 10 seconds, MLS2 must wait for three consecutive missed Hello messages before promoting itself to the Active role, a process that is independent of whether STP or RSTP is managing Layer 2 topology. The 12-second observed failover closely matches this expected 10–12 second detection window.

Reducing HSRP timers to sub-second values (for example, Hello 200 ms and hold 600 ms, as supported in IOS) would directly compress this window and could reduce device-failure failover to under 1 second, bringing all scenarios closer to the Five Nines availability target. This tuning was outside the scope of the present study and represents the highest-impact direction for future work. Shahriar and Fan [6] noted that HSRP achieved the lowest average time consumption and highest packet transmission rate among FHRP options in a multi-VLAN environment; sub-second timer tuning would further extend this advantage.

5.4 Implications for hospital network design

The baseline network, which lacked any FHRP, was entirely unrecoverable from a link or device failure without manual administrator intervention. In a hospital context, this translates directly to loss of access to Electronic Health Records, diagnostic

imaging systems and real-time clinical communication tools for the full duration of the outage. Healthcare organisations experience between four and six network downtime incidents annually, each averaging 2.5 to 4 hours [3]; a non-redundant gateway is incompatible with the availability demands of clinical operations.

The proposed enhancement — HSRP with RSTP and a MLS peer link — addresses this vulnerability non-disruptively. All original protocols (OSPF, DHCP, ACLs, NAT, PAT and IPsec VPN) were retained without modification, demonstrating that the redundancy layer can be introduced over an existing production configuration without service disruption. The incremental hardware cost is limited to the direct peer link between MLS1 and MLS2, making the solution economically accessible for hospitals that already operate a two-tier collapsed-core architecture. These properties align with the findings of Octavian [16] and Yogasetya Suhandi et al. [17], who demonstrated that HSRP-based redundancy can be integrated into existing enterprise topologies with minimal disruption, and extend those findings to a hospital-specific, multi-VLAN environment.

5.5 Limitations

All testing was conducted within Cisco Packet Tracer 9.0.0, a discrete-event simulator that approximates but does not fully replicate physical network behaviour. Hardware-specific factors including queuing delays, cable propagation characteristics and clock drift may affect failover timing in real deployments. Results should therefore be treated as indicative benchmarks rather than absolute production values.

Each failure scenario was evaluated over a single 100-second observation window with 50-packet sent, which may not capture variance across repeated trials or transient anomalies. HSRP timers were left at their default values throughout all tests; sub-second timer configurations were not evaluated. Security mechanisms including ACLs, NAT, PAT, SSH and IPsec VPN were not subjected to failure analysis since testing was confined to intra-site inter-VLAN traffic on the HQ side. These limitations define clear boundaries for interpreting the results and motivate the future work described in Section 6.

6. Conclusion

This study demonstrated that implementing HSRP and RSTP redundancy protocols in a simulated hospital network substantially enhances

its resilience and availability relative to a non-redundant baseline. The enhanced network, built upon the Medicine City Hospital design by Jassim [8], consistently outperformed the baseline across all tested failure scenarios. The most compelling result was recorded in Scenario 2, where the combination of HSRP, RSTP and a dedicated MLS–MLS peer link achieved approximately 99% availability, sub-second failover, zero packet loss and 100% PDR during a link failure event, which is an outcome that represents effectively transparent recovery from a perspective of end users.

Three principal conclusions emerge from this work. First, RSTP is essential whenever a peer link is present: without it, the additional topological complexity introduced by the peer link actively degrades STP-based configurations, worsening availability compared to a simpler single-uplink design. Second, HSRP timer expiry is the dominant bottleneck during device-level failures, and reducing these timers to sub-second values represent the most direct path to further availability improvement in that failure category. Third, the absence of any FHRP renders the network unrecoverable without manual intervention, confirming that gateway redundancy is a non-negotiable requirement for any hospital network design in which patient safety depends on continuous access to clinical systems.

Future work should prioritize physical hardware validation using real Cisco switches to characterize production failover behavior, sub-second HSRP timer tuning to address the device-failure bottleneck, and the addition of EtherChannel with LACP on access-layer uplinks to complement the gateway and Layer 2 redundancy provided by HSRP and RSTP. Adoption of a full three-tier architecture with a dedicated core layer would provide additional path diversity and support greater network scale. Evaluation against live application traffic such as simulated EHR queries or medical imaging transfers — rather than ICMP ping — would yield a more clinically meaningful measure of network performance under failure conditions. Together, these extensions would provide a comprehensive validation pathway toward the Five Nines availability standard that critical healthcare infrastructure demands.

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Conflict of Interest

The author has no conflicts of interest to declare.

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Author Contribution Statement

The author confirms sole responsibility for the following: study conception and design, network simulation and configuration, data collection, analysis and interpretation of results and manuscript preparation.

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